

This product is to be used only by a registered medical practitioner with experience in cardiology

EURYTHMIC Tablet 200 mg
Amiodarone Tablets BP 200 mg.

COMPOSITION:

Each uncoated tablet contains:
AMIODARONE HYDROCHLORIDE BP 200 mg.

DESCRIPTION:

White to off white colored, round, flat, uncoated beveled edge and having break line on one side of tablet.

Pharmacodynamics

Electrophysiology / Mechanism of Action: The antiarrhythmic effect of amiodarone may be due to at least two major properties:

- 1. Prolongation of myocardial cell-action potential duration and refractory period.
- 2. Noncompetitive α - and β -adrenergic inhibition.

Amiodarone prolongs the duration of the action potential of all cardiac fibers while causing minimal reduction of dV/dt (maximal upstroke velocity of the action potential). The refractory period is prolonged in all cardiac tissues without influencing resting membrane potential, except in automatic cells where the slope of the prepotential is reduced, generally reducing automaticity. These electrophysiologic effects are reflected in a decreased sinus rate of 15 to 20%, increased PR and QT intervals of about 10%, the development of U-waves, and changes in T-wave contour. These changes should not require discontinuation of amiodarone as they are evidence of its pharmacological action, although amiodarone can cause marked sinus bradycardia or sinus arrest and heart block. On rare occasions, QT prolongation has been associated with worsening of arrhythmia.

Hemodynamics: After intravenous administration, amiodarone relaxes vascular smooth muscle, reduces peripheral vascular resistance (afterload), and slightly increases cardiac index. After oral dosing, amiodarone produces no significant change in left ventricular ejection fraction (LVEF), even in patients with depressed LVEF. After acute intravenous dosing, amiodarone may have a mild negative inotropic effect.

Pharmacodynamics: There is no well-established relationship of plasma concentration to effectiveness, but it does appear that concentrations much below 1 mg/L are often ineffective and that levels above 2.5 mg/L are generally not needed. Some observations have suggested a plasma concentration, dose, or dose/duration relationship for side effects such as pulmonary fibrosis, liver-enzyme elevations, corneal deposits and facial pigmentation, peripheral neuropathy, gastrointestinal and central nervous system effects.

Monitoring Effectiveness: Predicting the effectiveness of any antiarrhythmic agent in long-term prevention of recurrent ventricular tachycardia and ventricular fibrillation is difficult and controversial, with highly qualified investigators recommending use of ambulatory monitoring, programmed electrical stimulation

(PES) with various stimulation regimens, or a combination of these, to assess response.

Several predictors of success not based on PES have also been suggested, including complete elimination of all nonsustained ventricular tachycardia on ambulatory monitoring and very low premature ventricular-beat rates (less than 1 VPB/1,000 normal beats).

Pharmacokinetics

Following oral administration, amiodarone is slowly and variably absorbed. The bioavailability of amiodarone is approximately 50%, (35 to 65%). Maximum plasma concentrations (C_{max}) are attained 3 to 7 hours after a single dose. Despite this, the onset of action may occur in 2 to 3 days, but more commonly takes 1 to 3 weeks, even with loading doses. Plasma concentrations with chronic dosing at 100 to 600 mg/day are approximately dose proportional, with a mean 0.5 mg/L increase for each 100 mg/day. These means, however, include considerable individual variability. Food increases the rate and extent of absorption of amiodarone. The area under the plasma concentration-time curve (AUC) and C_{max} of amiodarone increased by 2.3 (range 1.7 to 3.6) and 3.8 (range 2.7 to 4.4) times, respectively, in the presence of food. Food also increased the rate of absorption of amiodarone, decreasing the time to peak plasma concentration (T_{max}) by 37%. The mean AUC and mean C_{max} of desethylamiodarone increased by 55% (range 58 to 101%) and 32% (range 4 to 84%), respectively, but there was no change in the T_{max} in the presence of food.

Amiodarone has a very large but variable volume of distribution, averaging about 60 L/kg, because of extensive accumulation in various sites, especially adipose tissue and highly perfused organs, such as the liver, lung, and spleen. One major metabolite of amiodarone, desethylamiodarone (DEA), has been identified in man; it accumulates to an even greater extent in almost all tissues. DEA has significant electrophysiologic and antiarrhythmic effects generally similar to amiodarone itself. DEA's precise role and contribution to the antiarrhythmic activity of oral amiodarone are not certain. The development of maximal ventricular Class III effects after oral amiodarone administration in humans correlates more closely with DEA accumulation over time than with amiodarone accumulation.

Amiodarone is metabolized to DEA by the cytochrome P450 (CYP450) enzyme group, specifically cytochrome P450 3A4 (CYP3A4) and CYP2C8. The CYP3A4 isoenzyme is present in both the liver and intestines.

Amiodarone is eliminated primarily by hepatic metabolism and biliary excretion and there is negligible excretion of amiodarone or DEA in urine. Neither amiodarone nor DEA is dialyzable.

Age, sex, renal disease, and hepatic disease (cirrhosis) do not have marked effects on the disposition of amiodarone or DEA. Renal impairment does not influence the pharmacokinetics of amiodarone. Normal subjects over 65 years of age show lower clearances (about 100 mL/hr/kg) than younger subjects (about 150 mL/hr/kg) and an increase in t_{1/2} from about 20 to 47 days. In patients with severe left ventricular dysfunction, the pharmacokinetics of amiodarone are not significantly altered but the terminal disposition t_{1/2} of DEA is prolonged. Although no

dosage adjustment for patients with renal, hepatic, or cardiac abnormalities has been defined during chronic treatment with amiodarone, close clinical monitoring is prudent for elderly patients and those with severe left ventricular dysfunction.

Following single dose administration, amiodarone exhibited multi-compartmental pharmacokinetics with a mean apparent plasma terminal elimination half-life of 58 days (range 15 to 142 days) for amiodarone and 36 days (range 14 to 75 days) for the active metabolite (DEA). In patients, following discontinuation of chronic oral therapy, amiodarone has been shown to have a biphasic elimination with an initial one-half reduction of plasma levels after 2.5 to 10 days. A much slower terminal plasma-elimination phase shows a half-life of the parent compound ranging from 26 to 107 days, with a mean of approximately 53 days and most patients in the 40- to 55-day range. In the absence of a loading-dose period, steady-state plasma concentrations, at constant oral dosing, would therefore be reached between 130 and 535 days, with an average of 265 days. For the metabolite, the mean plasma-elimination half-life was approximately 61 days. These data probably reflect an initial elimination of drug from well-perfused tissue (the 2.5- to 10-day half-life phase), followed by a terminal phase representing extremely slow elimination from poorly perfused tissue compartments such as fat.

The considerable intersubject variation in both phases of elimination, as well as uncertainty as to what compartment is critical to drug effect, requires attention to individual responses once arrhythmia control is achieved with loading doses because the correct maintenance dose is determined, in part, by the elimination rates. Daily maintenance doses of amiodarone should be based on individual patient requirements.

Amiodarone and its metabolite have a limited transplacental transfer of approximately 10 to 50%. The parent drug and its metabolite have been detected in breast milk.

Amiodarone is highly protein-bound (approximately 96%). Consistent with the slow rate of elimination, antiarrhythmic effects persist for weeks or months after amiodarone is discontinued, but the time of recurrence is variable and unpredictable. In general, when the drug is resumed after recurrence of the arrhythmia, control is established relatively rapidly compared to the initial response, presumably because tissue stores were not wholly depleted at the time of recurrence.

INDICATION:

Treatment of severe rhythm disorders not responding to other therapy or when other treatments cannot be used: Atrial rhythm disorders (Conversion of fibrillation or flutter, and maintenance of sinus rhythm following conversion); nodal rhythm disorder; ventricular rhythm disorders (life-threatening ventricular premature contractions, ventricular tachycardia salvages, prevention of ventricular tachycardia attacks or ventricular fibrillation episodes); rhythm disorders associated with Wolff-Parkinson-White syndrome.

In view of its pharmaceutical properties, amiodarone is indicated in the previously mentioned rhythm disorders especially when they are associated with an underlying heart disease (coronary insufficiency, heart failure).

RECOMMENDED DOSE:

Initial Stabilization: Usual dosage is 600 mg daily and may be continued for 8-10 days.

Maintenance: The minimum effective dosage should be used according to individual response, it may range between 100-400 mg daily. Eurythmic may be given on alternate days (200 mg may be given every other day where 100 mg is recommended daily).

MODE OF ADMINISTRATION:

Oral

CONTRAINDICATION:

Amiodarone is contraindicated in patients with cardiogenic shock; severe sinus-node dysfunction, causing marked sinus bradycardia; second- or third-degree atrioventricular block; and when episodes of bradycardia have caused syncope (except when used in conjunction with a pacemaker). Amiodarone is contraindicated in patients with a known hypersensitivity to the drug or to any of its components, including iodine.

WARNING AND PRECAUTIONS:

WARNING: In the elderly, heart rate may be more markedly decreased

Amiodarone induces ECG changes: QT prolongation (in relation with prolonged depolarization) with the possible development of U-waves. These changes are evidence of its pharmacological action and do not reflect toxicity.

Amiodarone contains iodine: as a consequence, results of some thyroid function tests may be altered (radio-iodine uptake)

PRECAUTION:

Undesirable effects (see Adverse Reactions) are related in most cases with drug overloading, therefore careful attention should be paid in determining the minimum effective maintenance dosage in order to avoid or to minimize undesirable effects.

Patients should be instructed to avoid sun exposure or to use protective measures during therapy.

In patients with personal or family history of thyroid disorders, caution should be exercised when administering Amiodarone. Should it be required, the minimum effective dosage should be used and careful clinical and biological monitoring should be undertaken

Transplantation

In retrospective studies, amiodarone use in the transplant recipient prior to heart transplant has been associated with an increased risk of primary graft dysfunction (PGD).

PGD is a life-threatening complication of heart transplantation that presents as left, right or biventricular dysfunction occurring within the first 24 hours of transplant surgery for which there is no identifiable secondary cause (see section Adverse Effects). Severe PGD may be irreversible.

For patients who are on the heart transplant waiting list, consideration should be given to use an alternative antiarrhythmic drug as early as possible before transplant.

INTERACTIONS WITH OTHER MEDICATIONS

Combined Therapy with the Following Drugs That May Induce Torsades de Pointes is Contraindicated: Anti-arrhythmic agents

(bepridil, class Ia-quinidine-type agents, sotalol); non-anti-arrhythmic agents, including vincamine, sultopride, erythromycin IV, pentamidine in case of parenteral administration, as there is an increased risk of potentially lethal torsades de pointes.

Combined Therapy with the Following Drugs is not Recommended: β -blockers and certain calcium inhibitors (verapamil, diltiazem) as disturbances of automaticity (excessive bradycardia) and conduction may occur; stimulating laxative agents as their use may cause hypokalemia and therefore increase the risk of torsades de pointes (other of laxative agents should be used).

Caution should be exercised when using the following drugs in combination with Eurythmic:

Agents Which May Induce Hypokalemia: Diuretics inducing hypokalemia either alone or in combination, systemic corticosteroids (glycol-, mineralo-) tetracosactide and amphotericin B (IV).

It is necessary to prevent the occurrence of hypokalemia (and to correct hypokalemia when required). QT interval should be monitored and, in case of torsade de pointes, antiarrhythmic agents should not be given (electrolytic pacing should be instituted; IV magnesium may be used).

Oral Anticoagulant Agents: Due to the potentiation of oral anticoagulant therapy and the increased risk of bleeding, more frequent monitoring of prothrombin level and dosage adjustment of oral anticoagulant during treatment with and after discontinuation of amiodarone therapy is necessary.

Digitalis: Disturbances of automaticity (excessive bradycardia) and atrioventricular conduction (synergistic action) may occur. In addition with digoxin concentration (due to the decrease of digoxin clearance). Clinical ECG and biological monitoring (including digoxin plasma levels if relevant) should be undertaken. Dosage adjustment of digitalis treatment may be necessary.

Phenytoin: Possible increase of plasma phenytoin levels with signs of overdosage (particularly neurological signs). Clinical monitoring should be undertaken and phenytoin dosage should be reduced as soon as overdosage signs appear. Phenytoin plasma levels may be determined.

General Anaesthesia, Oxygen Therapy: Potentially severe complications have been reported in patients undergoing general anaesthesia: Bradycardia unresponsive to atropine, hypotension, disturbances of conduction, decreased cardiac output.

A few cases of severe respiratory complications, resulting sometimes in fatalities, have been observed most often in the period immediately after surgery (adult acute respiratory distress syndrome). A possible interaction with a high oxygen concentration may be implicated.

Before surgery, the anaesthetist should be informed that the patient is taking amiodarone.

Cyclosporine: Possible increase of cyclosporine plasma levels related to a decrease of the clearance of this drug. Dosages should be adjusted.

PREGNANCY AND LACTATION:

Pregnancy: Amiodarone can cause fetal harm when administered

to a pregnant woman. Although amiodarone use during pregnancy is uncommon, there have been a small number of published reports of congenital goiter/hypothyroidism and hyperthyroidism. If amiodarone is used during pregnancy, or if the patient becomes pregnant while taking amiodarone, the patient should be apprised of the potential hazard to the fetus. In general, amiodarone tablets should be used during pregnancy only if the potential benefit to the mother justifies the unknown risk to the fetus.

Lactation: Amiodarone and one of its major metabolites, desethylamiodarone (DEA), are excreted in human milk, suggesting that breast-feeding could expose the nursing infant to a significant dose of the drug. Therefore, when amiodarone therapy is indicated, the mother should be advised to discontinue nursing.

SIDE EFFECTS:

Tablet: Ophthalmological: Corneal microdeposits (almost always present), are usually limited to the area under the pupil and they do not require treatment discontinuation. They are rarely associated with coloured halos in dazzling light or sensation of fog. Corneal microdeposits consist of complex lipidic deposits and are reversible following treatment discontinuation.

Otherwise, a few cases of optic neuritis have been reported. At present, the relationship with amiodarone has not been formally established. If blurred or decreased vision occurs, ophthalmological examination is recommended.

Dermatological: Photosensitisation: Patients should be instructed to avoid sun exposure (UV rays exposure) during therapy. Cases of erythema have been also reported during the course of radiotherapy.

Skin rashes, usually of little specificity, including exceptional cases of exfoliative dermatitis have been reported. The relationship with the drug has not been formally established.

Slate grey or bluish pigmentations of the skin may occur in case of prolonged treatment with high daily dosages. Such pigmentations slowly disappear following treatment discontinuation.

In the absence of any clinical feature of thyroid dysfunction, "disassociated" biochemical changes are usual (increase in serum T4, T3 being normal or slightly decreased). Treatment should not be discontinued in such case.

Hypothyroidism: The diagnosis is supported by a clear increase in serum ultrasensitive TSH (TSHu). Euthyroidism is usually obtained within 1-3 months following treatment discontinuation. In life-threatening situations, amiodarone therapy could be continued and a supplementation with L-thyroxine being added, with TSH levels providing a guide to L-thyroxine dosage.

Hyperthyroidism: The diagnosis is supported by a clear decrease in serum. TSHu level. Amiodarone should be withdrawn. Clinical recovery usually occurs within 3-4 weeks following treatment withdrawal. Severe cases which may sometimes result in fatalities require emergency therapeutic management. The treatment should be adjusted to each individual case, antithyroid drugs which may not always be effective, corticosteroid therapy, β -blockers, etc.

The following clinical signs which are usually slight, should suggest the diagnosis of hypo- or hyperthyroidism: Weight gain, reduced activity, excessive bradycardia with regard to the expected effect of amiodarone, in case of hypothyroidism; weight loss, emergence of arrhythmia, angina, congestive heart failure in hyperthyroidism. The patient should be monitored several months following treatment discontinuation. Serum TSHu level should be measured if thyroid dysfunction is suspected. In patients at increased risk of thyroid dysfunction (family history of thyroid dysfunction, doubtful medical history), regular testing of thyroid function is recommended.

Pulmonary: Cases of pulmonary toxicity (hypersensitivity pneumonitis, alveolar/interstitial pneumonitis or fibrosis, pleuritis, bronchiolitis obliterans with organizing pneumonia (BOOP), resulting sometimes in fatalities, have been reported. Chest x-ray should be performed in patients developing dyspnea (at effort) whether in isolation or associated with deterioration of general health status (fatigue, weight loss, fever). The pulmonary disorders are generally reversible following early withdrawal of amiodarone therapy. Corticosteroid therapy may also be considered. Clinical signs usually resolve within 3-4 weeks, followed by radiological and lung function improvement (several months).

A few cases of adult acute respiratory distress syndrome, resulting sometimes in fatalities have been observed, most often immediately after surgery (a possible interaction with a high oxygen concentration may be implicated).

Neurological: Sensitive motor peripheral neuropathy and/or myopathy, usually reversible on withdrawal of the drug.

Others: Extrapramidal tremor, cerebellar ataxia, exceptional benign intracranial hypertension (cerebral pseudotumor, nightmares).

Hepatic: Regular monitoring of liver function test (transaminases) is recommended during treatment. Isolated elevation of serum transaminases which are usually moderate (1.5-3 times normal) have been reported at the beginning of therapy. They may regress with dose reduction or even spontaneously. A few cases of acute liver disorders with elevated serum transaminases and/or jaundice, which included some fatalities, have been also reported. In such cases, treatment should be discontinued.

There have also been reports of chronic liver disease (pseudoalcoholic hepatitis, cirrhosis). Clinical signs and biological changes may be minimal (possible hepatomegaly, transaminases elevated 1.5-5 times normal); therefore, regular monitoring of liver function tests is recommended during therapy. Clinical and biological abnormalities usually regress upon cessation of treatment. Nevertheless, fatal cases have been reported.

Cardiac: Bradycardia which is generally moderate and dose-related may occur. In some cases (sinus node dysfunction, elderly patients), marked bradycardia or more exceptionally, sinus arrest have been reported.

There have been rare instances of conduction disturbance (sinoatrial block, AV block of various degrees).

Occurrence of new arrhythmia or aggravation of the preexisting trouble, followed in some cases by cardiac arrest, have been reported. In view of current knowledge, it is not possible to differentiate what may be due to the drug from what may be related to the underlying cardiac conditions or may be due to a lack of efficacy of therapy. These effects are more rarely reported than with most of the other anti-arrhythmic agents and they occur in general in case of certain drug interactions or electrolyte imbalance (see Interactions). These reactions were also observed with the IV route.

Injury, poisoning and procedural complications: Frequency 'not known': Potentially fatal primary graft dysfunction post cardiac transplant (See section Warnings and Precautions).

Others: Benign digestive disorders (nausea, vomiting, dysgeusia) which usually occur with loading dosage and which regress with dose reduction; alopecia; a few cases of epididymitis have been reported (the relationship with the drug has not been established); there have been rare cases of various clinical features which may suggest hypersensitive reactions: Vasculitis, renal involvement with elevation of creatinine levels, thrombocytopenia.

A few exceptional cases of haemolytic anemia or aplastic anemia have also been reported.

SYMPTOMS AND TREATMENT OF OVERDOSE:

There have been cases, some fatal, of amiodarone overdose. In addition to general supportive measures, the patient's cardiac rhythm and blood pressure should be monitored, and if bradycardia ensues, a β -adrenergic agonist or a pacemaker may be used. Hypotension with inadequate tissue perfusion should be treated with positive inotropic and/or vasopressor agents. Neither amiodarone nor its metabolite is dialyzable.

STORAGE CONDITION:

Store below 30°C, protected from light.

SHELF LIFE:

3 years

PACK SIZE:

10x10's strip pack.

DATE OF REVISION:

28/11/2025

Product Registration Holder:

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